OCNG 470 Lab 2 Assignment

Probability and Probability Distributions in Python

Lab Objectives

By the end of this lab, you should be able to:

1. Calculate and plot a **binomial probability distribution** in Python.
2. Calculate and plot a **Poisson probability distribution** in Python.
3. Calculate and plot a **normal/Gaussian probability distribution** in Python.
4. Visualize probability distributions as bar charts.
5. Use Python functions and packages including `math.factorial`, `numpy`, `scipy.stats`, `matplotlib.pyplot`, and `for` loops.

Lab Activities

There are three parts to this assignment. Complete all parts.

Each part is worth **10 points**.

- **Part 2A:** Binomial Distribution
- **Part 2B:** Poisson and Normal Distributions
- **Part 2C:** Binomial Probability with a Four-Sided Die

Due Date

The due date is listed on Canvas under the **Assignments** tab for your lab section.

Lab Outcomes

You will submit:

1. One Jupyter Notebook file: `.ipynb`
2. One exported PDF document

Submit both files to Canvas under the **Assignments** tab.

Submission Instructions

1. Complete the assignment in a new Jupyter Notebook.
2. Save your notebook using the following naming convention:

`OCNG470_LabXX_Fall2024_SectionYYY_ZZ.ipynb`

where:

- XX = lab number, such as Lab02
- YYY = section number, such as 500 or 502
- ZZ = your initials

3. At the beginning of your notebook, include the following information in a Markdown cell:

Name: Your Name

Date: MM-DD-YYYY

UIN: XXXXXXXXX

Filename: OCNG470_LabXX_Fall2024_SectionYYY_ZZ.ipynb

Purpose: To learn about binomial, Poisson, and normal probability distributions in Python

4. For questions that require written answers, type your responses in Markdown cells below your code.
5. Before submitting, make sure your notebook runs from beginning to end without error messages.
6. Make sure all figures have clear titles and axis labels.

Required Python Packages

Run the following code cell at the beginning of your notebook:

```
import numpy as np

import matplotlib.pyplot as plt

from scipy.stats import binom, poisson

from math import factorial, exp, sqrt, pi
```

Part 2A: Binomial Distribution

10 points

In this section, you will model the number of heads obtained from tossing a coin 25 times.

Use the following parameters:

n **=** 25

p **=** 0.5

where:

- n = number of independent Bernoulli trials
- p = probability of success
- X = number of successful outcomes
- P(X) = probability of obtaining X successful outcomes

For this problem, a “success” means getting heads.

The binomial probability equation is:

$$P(X=k)=\frac{n!}{k!(n-k)!}p^{k}(1-p)^{n-k}$$

Step 1: Start Part 2A

Add a Markdown heading in your notebook:

#Part 2A. Binomial Distribution

Step 2: Calculate the probability of getting exactly 7 heads

Using factorial() and power operations, calculate the probability of getting exactly 7 heads after tossing a coin 25 times.

For this problem:

n **=** 25

p = 0.5

k = 7

Calculate:

$$P(X=7)$$

$$P(X=7)=\frac{25!}{7!(25-7)!}(0.5)^{7}(1-0.5)^{25-7}$$

Use the following code structure:

n = 25

p = 0.5

k = 7

Calculate P(X = 7) manually using factorial() and powers

```
manual_prob = factorial(n) / (factorial(k) * factorial(n-
k)) * p**k * (1-p)**(n-k)
```

Check your answer using scipy.stats.binom.pmf()

```
python_prob = binom.pmf(k, n, p)
```

```
manual_prob, python_prob
```

In a Markdown cell below your code, answer:

The probability of getting exactly 7 heads in 25 tosses is ...

Your manually calculated probability should match the output from:

```
binom.pmf(7, n, p)
```

Step 3: Calculate and plot the full binomial distribution for a fair coin

Calculate probabilities for all possible values of X from 0 to 25.

```
X = np.arange(0, 26)
```

```
pdf_fair = binom.pmf(X, n, p)


X, pdf_fair
```

Plot the binomial probability distribution as the first subplot in Figure 1.

```
plt.figure(figsize=(10, 8))


plt.subplot(2, 1, 1)
plt.bar(X, pdf_fair)
plt.xlabel("Number of Heads")
plt.ylabel("Probability")
plt.title("Binomial Distribution: n = 25, p = 0.5")
plt.xticks(X)


plt.tight_layout()
plt.show()
```

Step 4: Describe the fair coin distribution

In a Markdown cell, describe the shape of the distribution.

Address the following questions:

- Where is the distribution centered?
- Is it symmetric or skewed?
- What is the approximate mean?
- Why does the distribution have this shape?

For a binomial distribution, the mean is:

$$\mu = np$$

For the fair coin:

$$\mu = 25(0.5) = 12.5$$

Step 5: Repeat with a biased coin

Now repeat the calculation using a biased coin where the probability of heads is:

```
p_biased = 0.8
```

Calculate the full distribution for X = 0 to 25.

```
p_biased = 0.8
pdf_biased = binom.pmf(X, n, p_biased)


X, pdf_biased
```

Plot the fair coin and biased coin distributions as two subplots in the same figure.

```
plt.figure(figsize=(10, 8))


plt.subplot(2, 1, 1)
plt.bar(X, pdf_fair)
plt.xlabel("Number of Heads")
plt.ylabel("Probability")
plt.title("Binomial Distribution: n = 25, p = 0.5")
plt.xticks(X)


plt.subplot(2, 1, 2)
plt.bar(X, pdf_biased)
plt.xlabel("Number of Heads")
plt.ylabel("Probability")
plt.title("Binomial Distribution: n = 25, p = 0.8")
plt.xticks(X)
```

```
plt.tight_layout()
plt.show()
```

Step 6: Compare the fair and biased coin distributions

In a Markdown cell, answer the following questions:

- How does the biased coin distribution differ from the fair coin distribution?
- Did the center of the distribution move?
- Did the shape change?
- Why did the probability distribution change when p changed from 0.5 to 0.8?

For the biased coin, the mean is:

$$\mu = np$$

$$\mu = 25(0.8) = 20$$

Part 2B: Poisson and Normal Distributions

10 points

In this section, you will calculate and plot Poisson probability distributions. Then you will calculate and plot normal distributions and compare them to the Poisson distributions.

Use Poisson parameters:

$$\lambda = 1,4,10,15$$

You will calculate the probability of observing:

$$x = 4$$

for each value of λ.

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

- x = number of observed occurrences
- λ= average number of occurrences per sample

- e= exponential constant, approximately 2.7183

Step 1: Start Part 2B

Add a Markdown heading in your notebook:

Part 2B. Poisson and Normal Distributions

Step 2: Define the Poisson parameters

Use the following λ values:

```
Lambda = [1, 4, 10, 15]
x = 4
```

Step 3: Calculate $P(x=4)$ for each λ manually

Using factorial(), powers, and exp(), calculate the probability that a single sample contains exactly four observations when:

lambda = 1

lambda = 4

lambda = 10

lambda = 15

Use a for loop.

```
Lambda = [1, 4, 10, 15]
x = 4


for lam in Lambda:
    # Calculate Poisson probability manually
    calc_prob = lam**x * exp(-lam) / factorial(x)


# Check with scipy.stats.poisson.pmf()
```

```
scipy_prob = poisson.pmf(x, lam)


print(f"P(x = 4, lambda = {lam}) manually = {calc_prob}")
print(f"P(x = 4, lambda = {lam}) scipy    = {scipy_prob}")
print()
```

In a Markdown cell below your code, report your answers:

```
P(x = 4, lambda = 1) = ...
P(x = 4, lambda = 4) = ...
P(x = 4, lambda = 10) = ...
P(x = 4, lambda = 15) = ...
```

Step 4: Plot four Poisson distributions

Calculate the full Poisson distribution for:

```
X = np.arange(0, 31)
```

Use `poisson.pmf()` to calculate probabilities for each λ value.

Plot the four Poisson distributions as subplots in Figure 2 using a 2-by-2 layout.

```
X = np.arange(0, 31)
Lambda = [1, 4, 10, 15]
plt.figure(figsize=(12, 8))
for i, lam in enumerate(Lambda):
    poisson_pdf = poisson.pmf(X, lam)
    plt.subplot(2, 2, i + 1)
    plt.bar(X, poisson_pdf)
    plt.xlabel("X")
    plt.ylabel("Probability")
    plt.title(f"Poisson Distribution: lambda = {lam}")
```

```
plt.tight_layout()
plt.show()
```

Step 5: Describe the Poisson distributions

In a Markdown cell, describe how the distributions change as λincreases.

Address the following questions:

- How does the center of the distribution change?
- How does the spread change?
- How does the skewness change?
- What is the relationship between $\lambda$and the mean of a Poisson distribution?

For a Poisson distribution:

$$\text{mean} = \lambda$$

and

$$\text{variance} = \lambda$$

Step 6: Plot two normal distributions

Now plot two normal distributions for:

```
X = np.arange(0, 101)
```

Use the following parameters:

```
mu = 25
sigma = [5, 10]
```

The normal probability density function is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

For this lab, write the equation manually instead of using a built-in normal distribution function.

```
X = np.arange(0, 101)

mu = 25

sigma_values = [5, 10]

plt.figure(figsize=(10, 8))


for i, sigma in enumerate(sigma_values):

        # Calculate normal PDF manually

    normal_pdf = (1 / (sigma * np.sqrt(2 * np.pi))) *
np.exp(-0.5 * ((X - mu) / sigma)**2)

    plt.subplot(2, 1, i + 1)

    plt.bar(X, normal_pdf)

    plt.xlabel("X")

    plt.ylabel("Probability Density")

    plt.title(f"Normal Distribution: mu = {mu}, sigma =
{sigma}")


plt.tight_layout()

plt.show()
```

Step 7: Describe the normal distributions

In a Markdown cell, answer the following questions:

- How are the two normal distributions different from each other?
- Which distribution is taller and narrower?
- Which distribution is shorter and wider?
- Why does changing σaffect the shape of the distribution?
- How are the normal distributions different from the Poisson distributions?

Part 2C: Binomial Probability with a Four-Sided Die

10 points

In this section, you will calculate probabilities for rolling a four-sided die five times.

A four-sided die has four possible outcomes:

1, 2, 3, 4

The probability of rolling a "1" on any single roll is:

$$p = \frac{1}{4}$$

The probability of not rolling a "1" is:

$$1 - p = \frac{3}{4}$$

Step 1: Start Part 2C

Add a Markdown heading in your notebook:

> **# Part 2C. Binomial Probability with a Four-Sided Die**

Step 2: Probability of rolling exactly two "1"s in five rolls

Calculate the probability of rolling a four-sided die five times and getting exactly two "1"s.

Use:

```
n = 5

p = 1/4

k = 2
```

Calculate the probability using `binom.pmf()`.

```
n = 5

p = 1/4

k = 2


prob_two_ones = binom.pmf(k, n, p)
```

```
prob_two_ones
```

Also show the calculation manually using the binomial probability equation:

$$P(X=2)=\binom{5}{2}\left(\frac{1}{4}\right)^2\left(\frac{3}{4}\right)^3$$

```
manual_two_ones = factorial(n) / (factorial(k) * factorial(n-k)) * p**k * (1-p)**(n-k)
```

```
manual_two_ones
```

In a Markdown cell, report your answer:

The probability of rolling exactly two "1"s in five rolls is ...

Step 3: Probability of rolling exactly two consecutive "1"s

Now calculate the probability of rolling a four-sided die five times and obtaining exactly two "1"s, where the two "1"s occur consecutively.

Important: This is not the same as simply rolling two "1"s anywhere in five rolls. Because the two "1"s must be next to each other, you must calculate this manually instead of using binom.pmf().

The possible positions for the consecutive pair are:

11xxx

x11xx

xx11x

xxx11

where x means “not 1.”

There are four possible positions for the consecutive pair.

For each pattern:

- probability of rolling "1" twice is:

$$\left(\frac{1}{4}\right)^2$$

- probability of not rolling "1" in the other three positions is:

$$\left(\frac{3}{4}\right)^3$$

Therefore, the probability is:

$$4 \times \left(\frac{1}{4}\right)^2 \times \left(\frac{3}{4}\right)^3$$

Write this calculation in Python:

```
p_one = 1/4

p_not_one = 3/4

num_consecutive_positions = 4

prob_consecutive = num_consecutive_positions * p_one**2 * p_not_one**3

prob_consecutive
```

In a Markdown cell, report your answer:

The probability of rolling exactly two consecutive "1"s in five rolls is ...

Then explain why this probability is different from the probability in Step 2.

Final Checklist Before Submission

Before submitting, make sure:

- Your .ipynb notebook runs without errors.
- Your notebook includes your name, date, UIN, filename, and purpose.
- Each part has a clear Markdown heading.
- All written answers are included in Markdown cells.
- All figures have clear titles and axis labels.
- Your PDF displays all figures correctly.
- You submit both the .ipynb file and the exported PDF to Canvas.